

Ammar Ahmed

Mechanical Work Samples (2-3 Projects)

Mechanical design and prototyping experience spanning UAV structures, propulsion concepts, and rapid manufacturing. Focused on CAD-driven development, lightweight structures, and integration of mechanical assemblies with electronics and test setups.

UAV Design, Fabrication & Integration

This project focused on the mechanical design and prototyping of a foldable fixed-wing UAV. I developed the airframe architecture, including fuselage structure, wing mechanisms, and mounting interfaces for avionics and payloads. Multiple prototype iterations were produced using FDM printing and rapid assembly techniques, allowing quick evaluation of stiffness, packaging, and serviceability. The work emphasized manufacturability, lightweight structures, and reliable integration with flight-control and sensor hardware. The resulting prototypes supported aerodynamic testing and system-level integration prior to flight trials.

My Role / Contributions

- Led structural design of a foldable fixed-wing UAV with manufacturable assemblies and serviceable subsystems.
- Built and validated 3D-printed airframe prototypes; iterated on ribs, spars, and fuselage joints for stiffness and weight.
- Integrated mechanical components with flight electronics, sensors, and actuation hardware for test-ready builds.
- Supported fabrication and assembly workflows to transition prototypes toward flight-ready configurations.

Tools/Skills: SolidWorks, FDM 3D printing (PLA/ABS), assembly tooling, basic CFD, systems integration



Meshed geometry of a UAV



3D Printed Body of UAV before assembly



3D printed and assembled Fixed-wing Foldable UAVs

Conceptual Development of Launch Vehicle and Rocket Engines

I contributed to early-stage mechanical design of launch-vehicle propulsion hardware. The work involved creating CAD concepts for injector, chamber, and nozzle geometries, with emphasis on clear mechanical interfaces and integration constraints. I generated renders and cutaway views that communicated flow paths, mounting features, and component relationships to stakeholders. The concepts balanced geometric feasibility with manufacturability, supporting downstream decisions on fabrication approach and subsystem packaging.

My Role / Contributions

- Developed conceptual CAD layouts for engine injector, chamber, and nozzle assemblies.
- Produced renders and section views to communicate flow paths and interface geometry.
- Considered manufacturability and integration constraints across propulsion subsystem interfaces.

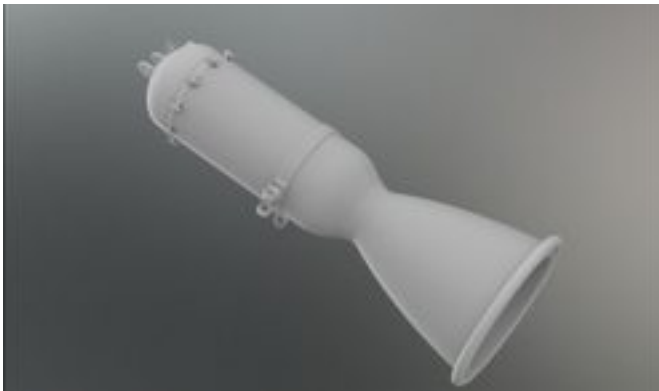
Tools/Skills: SolidWorks/NX, mechanical CAD, design for manufacturability, visualization rendering



Launch vehicle render



Rocket engine concept



Rocket engine concept detail

3D Printing

This body of work highlights mechanical prototyping through FDM 3D printing. I prepared CAD models for print by setting tolerances, split lines, and support strategies, then produced parts to validate fit, motion, and assembly interfaces. The prototypes were used to assess structural behavior and refine designs quickly without tooling lead time. Post-processing and re-prints were used to converge on manufacturable geometries suitable for integration into larger assemblies.

My Role / Contributions

- Prepared CAD for FDM printing with proper tolerances, part orientation, and support strategy.
- Produced functional prototypes and assemblies to validate fit, motion, and structural behavior.
- Performed post-processing and iterative refinements based on test feedback.

Tools/Skills: FDM 3D printing, CAD pre-processing, rapid prototyping, tolerance tuning



3D printed part close-up



3D printed part assembly